cnc_app User Manual

Oct 31, 2026 @deltazerorsan

1. What it is and how it works

cnc_app converts a vector drawing (an SVG exported from a vector drawing application, or one drawn inside cnc_app itself) into G-code for a CNC machine running GRBL 1.1, ready to load into UGS. It handles 2D work: through cuts, pockets, drilling and lines. This manual assumes you already know how to operate a CNC machine and only explains how the application handles each step.

Fixed rules:

- **Z = 0 is the top surface of the material.** A through cut plunges down to −(thickness + margin).
- **The spindle only switches on and off**, with no speed control. The feed rate comes from the tool profile.
- **Insides (holes) are cut before outsides (parts)**, so a part doesn't come loose too early.
- **A rectangle with shapes inside is a board**: the stock material (the sheet). It isn't cut, its bottom-left corner is zero, and it gets its own `.nc` file.
- **The application does not send the G-code to the machine**, doesn't home, doesn't set zero and doesn't simulate the cut: that is UGS's job (Universal G-code Sender).

The job, from start to finish: pick the machine (if there is more than one), open the SVG or start an empty drawing, pick the tool and the thickness, review the operations table, check the toolpath in the *Machine* view, generate the `.nc` and load it into UGS. If needed, the *Hold-down* tab suggests where to nail the sheet down.

2. Installation

No installation is needed: copy the application with its data files into a folder (a USB stick works too) and open `cnc_app.exe`. The data files always sit next to the `.exe`:

- `maquinas.json`: the machine profiles. Without it the application starts but generates nothing.
- `herramientas.json`: the tool profiles.
- `cnc_app.toml`: default values (automatic rotation, 0.4 mm margin...). Read at startup.
- Recipes (`.toml`), if you use them.

Tool profiles are re-read with *File → Reload tool profiles* (F5) and machine profiles with *File → Reload machines*. Each PC remembers on its own the window position, the last folder used, the job and G-code folders (section 12) and the tabs switch: none of that travels with the folder.

3. The window

Area	What is there
Menu bar	*File*, *Edit*, *Recipe*, *Shapes*, *Draw*, *View*, *Help*
Left column	The *Boards* list, the *Detect boards* checkbox and the job settings (section 5)

Area	What is there
Center	The *Drawing*, *Machine*, *Rapids* and *Fit* buttons; *Generate G-code…*; the mouse tool strip and the canvas
Bottom tabs	*Operations*, *Messages* and *Hold-down*
Status bar	Tells you what to do next at each moment

The two views (Ctrl+1 and Ctrl+2). *Drawing* shows the SVG as it is, with the machine travel as blue dots and a red cross where zero will land, so you can see whether the board fits. *Machine* shows the toolpath already placed and rotated, each operation in its own color; *Rapids* adds the G0 moves between cuts. On the canvas, the wheel zooms (*View → Reverse mouse wheel* flips the direction), dragging pans the view, and *Fit* (Ctrl+0) frames everything. When you switch between *Drawing* and *Machine* (in either direction), the zoom level and the area you are looking at are kept. The *Grid* and *Travel* buttons toggle on and off, and the PC remembers how you left them.

Messages is where the application flags anything that doesn't fit: shapes without an operation, toolpaths that run off the board (with a button to apply the suggested shift), groups discarded when a job is opened… If something doesn't come out as expected, look there first. Every change is recalculated in the background.

Language. The application appears in Spanish or English, depending on the Windows language; change it in *Help → Language* and it takes effect the next time the application starts. The G-code is the same in both: only its comments change. Numeric boxes always accept either a comma or a point as the decimal separator.

4. Opening a drawing

There are three ways to start: *File → Open SVG drawing…* (Ctrl+O) or drop the SVG onto the window; *File → Open saved job…* (Ctrl+Shift+O) to resume a `.cncjob`; and *Draw → New empty drawing* (Ctrl+N) to draw from scratch. There is only one window: opening a file while the application is already running loads it into that same window.

What it reads. Each path is a shape with its own stroke color and fill color. Text and images are ignored (in the drawing application, convert text to curves). Shapes with no area, duplicated shapes and shapes that cross themselves are flagged in *Messages*.

Color rules. The operations table has one row for each stroke/fill combination. Until you change anything, everything is a normal through cut: the designer uses color to mark what each thing is, and you tell the application what each color does (section 6).

Where zero lands. Measurements are in mm, with X to the right and Y up. With a board, zero is its bottom-left corner. With no board, the material is assumed to sit at the bottom-left corner of what is cut, and zero lands 4 mm outside it (change this in *Zero at (mm) from drawing*). Anything outside every board is not cut, and you get a warning.

Rotation. *Automatic* (the default) rotates the job 90° only if it doesn't fit as it is within the machine travel but does fit rotated. The other options are *No rotation*, *90° counterclockwise*, *180°*, *270° (90° clockwise)* and *Vertical* (long side along Y). If it doesn't fit any way, nothing is generated.

5. Job settings

They live in the panel in the left column. Anything you leave empty is taken from the tool profile (shown in gray). If a value isn't anywhere, the application doesn't invent one: it stops and tells you which one is missing. **Priority of each value:** the row of the operations table, then the job setting, then the tool profile.

Group	Field	What it does
Tool and material	Tool	A profile, or a diameter in mm typed by hand. *Profiles…* opens the editor
	Thickness (mm)	Required. A through cut plunges to thickness + margin
	Spoilboard margin (mm)	How far to cut into the spoilboard beyond the material (0.4 by default)
	Zero at (mm) from drawing	Only with no board: how many mm outside the bottom-left corner zero sits
	Rotation	Automatic, No rotation, 90° counterclockwise, 180°, 270° (90° clockwise) or Vertical
Feeds	Cut (mm/min), Plunge (mm/min), Pass depth (mm)	Override the tool profile's values for the whole job
G-code	Safe Z (mm) / Approach Z (mm)	Height of the free moves (5 mm) and the height from which plunging starts (1 mm)
	Switch the spindle on and off (M3 / M5)	No speed value: the VFD (the spindle speed controller) has to be set by hand
	Pause (M0) before starting the spindle	The machine stops until you press *Resume* in UGS
	Home ($H) at the end / Return to origin at the end	Add a Home or a return to X0 Y0 at the end
	End code	Your own G-code, one command per line, placed before M30
Board	Generate this board	The board gets its own `.nc`
	Thickness (mm), Margin (mm), Rotation	Only if this board is a different material; empty = the general value
	Shift drawing (mm)	Moves the drawing inside the board
Tabs	Use tabs	Master switch (section 10)
	Width, Height, One every (mm of	6 mm, 2 mm, and one tab every 150 mm of perimeter

Group	Field	What it does
	perimeter)	

With neither Home nor return to origin, the G-code ends by raising Z, stopping the spindle and staying wherever it finishes.

The *Board* group applies to the board selected in the list on the left. If the toolpath runs off the board, *Messages* tells you by how much and on which side, and offers to apply the shift that fixes it, with 4 mm of clearance.

6. Operations table

This is the *Operations* tab. Each row is a stroke/fill combination (or a group, section 8) and says what to do with the shapes of that color. Active rows are cut **in table order**, which should always be drills, pockets, insides, lines and outsides; the table maintains this order by itself. The and buttons move a row by hand, and if the order stops being correct a red warning appears and *Sort* fixes it. If everything is a normal through cut, you don't need to touch anything.

Column	What it is
Color	Stroke and fill of the shapes in that row
Shapes	How many shapes fall in the row
Operation	*Cut*, *Pocket*, *Drill* or *Ignore*
Side	*Cut* only: *Auto*, *Outside*, *Inside* or *On the line*
Through	Cuts down to thickness + margin
Tabs	Leaves tabs on the through cuts of that row
Depth (mm)	Total depth, if not a through cut
Pass depth (mm)	Depth of each pass (for a drill, the peck)
Feed, Plunge	Cutting feed and plunge feed, in mm/min
Stepover %	Pocket stepover, as a % of the tool diameter (40, not 0.4)

Empty boxes take the value from the job settings or the tool profile.

- **Cut** follows the outline. With *Auto*, a shape inside another is a hole (the tool runs on the inside) and a shape outside is a part (the tool runs on the outside). *On the line* puts the tool right on the stroke and accepts open lines. With no depth given, it is a through cut.
- **Pocket** removes all the material inside the shape down to the given depth (that area is a pocket). Shapes of the same color inside another one are **islands** and are left untouched. The smaller the stepover, the more laps: a large pocket can go from 5 h to 30 h of machine time when the stepover drops from 60 % to 10 %, and the "≈ … of machine time" line in the summary gives it away before you generate.
- **Drill** makes one plunge at the center of each circle. The hole size comes from the tool diameter; if the circle has a different diameter, the application warns you. For larger holes, use *Pocket*.

- **Ignore** doesn't cut: use it for guides, dimensions and notes. In the *Drawing* view they appear dimmed.

Recipes. The table is saved as a `.toml` file (*Recipe* → *Save recipe as…*) and reopened with *Open recipe…*. *Reset table* sets every row back to a normal through cut. Recipe colors are written as `#rrggbb`; a badly written color is an error, not silently taken as black.

7. Tools and machines

Tools. Each tool is a profile. Pick it in the *Tool* selector and edit it with *Profiles…* (or *File* → *Tool profiles…*, Ctrl+T). The editor has the list on the left and the form on the right, with *New*, *Duplicate* (for another material) and *Delete*. An invalid profile is marked with and prevents saving; when you save, a `.bak` copy of the previous file is kept.

Field	Meaning
Diameter (mm)	The only required field
Type	*Flat end mill* (default), *Ball nose* or *V-bit*; a V-bit also needs *V angle (°)*
Flutes, Useful length (mm)	Number of cutting edges, and mm of usable flute: the application refuses to cut deeper than the flute
Cutting feed (mm/min), Plunge feed (mm/min)	Cutting and plunging speeds
Pass depth (mm)	How deep each pass goes. If empty, the cut is made in a single pass
Stepover (% of diameter)	Distance between passes, used for pocketing
RPM, Name, Notes	Informational only

Only flat end mills generate toolpaths; ball nose and V-bit profiles are stored as data. If there is no cutting feed in either the profile or the job settings, the application stops and says so: it never cuts with an invented value. The *Identifier* of a new profile must start with a letter (`em4_mdf`), because a bare number is taken as a diameter.

Machines. Each job carries **its own machine**: a profile with the travel (X, Y, Z) and whatever is known about the controller. Pick it in the *Machine* selector (the last one used is remembered on each PC), since the PC where you draw is not necessarily the one that drives the machine. To create a profile: in the UGS console, right after connecting, type `$$` and `$I`, copy everything and paste it into *Machines…* (Ctrl+M) → *New from a log…*. The application shows what it understood, what it could not read and, if the profile already existed, what would change; it is only accepted if the log includes the travel.

Anything missing counts as "unknown": only the travel is required, and a missing value is not checked. With what the profile knows, the application **will NOT write the G-code** if:

- the toolpath doesn't fit in X or Y;
- the depth plus the safe Z don't fit in Z;
- Home (`$H`) at the end is ticked and the machine profile has no Homing (`$22`) (this only happens with an emulator, which can't home; a real machine always has it);
- the spindle is started on a machine in laser mode (`$32=1`).

A feed higher than the profile's maximum is only a warning. The `.cncjob` stores a copy of the machine profile; when you open it, if the profile has changed or no longer exists, *Messages* tells you. The second line of the `.nc` carries `(Machine: name | travel…)`, so you don't take another machine's G-code to the shop. With no `maquinas.json`, or with several machines and none chosen, nothing is generated.

8. Boards and groups

Boards. One SVG can bring several materials at once. There is a single convention: **a rectangle with shapes inside is a board**. Each board is a job of its own: it comes out in its own `.nc` and has its zero at its bottom-left corner; the rectangle itself is not cut.

- A shape belongs to a board if at least half of it falls inside. Anything outside every board is not cut, and you get a warning.
- A rectangle inside a board is an ordinary part. A circle or ellipse that contains boards only groups them and is discarded.
- Boards are numbered left to right by columns, and top to bottom within each column.
- A rectangular plate with holes is indistinguishable from a board with parts: untick *Detect boards* and it will be cut as a part.

The *Boards* list has a checkbox per board; when you select one, the *Board* group in the settings gives it its own thickness, margin, rotation and shift. *Generate G-code…* writes one `.nc` per ticked board (`NAME_board1.nc`, …). Boards keep their values and hold-down points when you move or copy shapes, and the points follow the board if you move it.

Selecting. This is for when the design doesn't distinguish operations by color (everything black, but with pockets): instead of asking the designer for another color, you select the shapes by hand and give them a row of their own in the table. Selection is done in the *Drawing* view: a click on a line selects a shape; Ctrl+click adds or removes a shape; Shift+drag draws a box and selects the shapes that fit entirely inside it; *Edit* → *Select all* (Ctrl+A) takes parts and holes, never the board frame; Esc clears the selection. The *Select* button above the table selects by nesting level (0 is the outsides, 1 what is inside another shape, 2 what is inside that…) or the open lines.

Groups. Once you have a selection, *Move to group* → *New group…* (Ctrl+Shift+N) asks for a name and a type: *Pocket*, *Cut*, *Drill* or *Ignore*. A new row appears, placed where the row order requires, and there you set depth and feeds as in any other row. *Remove from group* (Ctrl+Shift+U) returns the shapes to their color; a group with no shapes disappears. Groups are saved in the job (if the drawing changes when you open it, they are discarded and you are warned), but a recipe does not carry them.

Separate inside/outside. A cut row set to *Auto* cuts correctly, but it is a single operation. *Separate inside/outside* (Ctrl+Shift+I) splits it into three rows (*Insides*, *Lines* and *Outsides*) with the same settings: the G-code comes out the same, but each row can now have its own depth, feed or pass depth. Undo with Ctrl+Z.

Row order. The correct order is always **drills, pockets, insides, lines and outsides**: small and inner things first, so the part comes loose last. The application respects this order when you open a drawing, create a group, or change the type or side of a row. If you move rows by hand and the order stops being correct, a red warning appears (also in *Messages* and in the summary) and *Sort* fixes it; if an *Auto* row mixes holes and parts, the warning asks you to separate it first.

9. Drawing and editing

This is for adding or touching up parts without going back to the vector drawing application. *Draw* → *New empty drawing* (Ctrl+N) opens a 240 × 160 mm page with zero marked by a red cross; a new drawing has no SVG and everything is saved in the job. Measurements are in mm (comma or point), with X to the right and Y up.

With the mouse (*Drawing* view). The tool strip above the canvas, or a key, picks what to draw. The status bar says what to do at each step and, next to the cursor, a label shows the live measurements.

Tool	Key	How it works
Select	V	Select shapes, and drag to pan the view
Rectangle / Ellipse	R / E	Two corners of its box
Circle	C	The center and a point on the edge
Polygon	P	The center and a vertex (this also sets the rotation)
Slot	S	The centers of its two ends
Line	L	Point by point; finish with a double-click, Enter or right-click; clicking the first point closes the shape
Move	M	A start point and an end point, applied to the selection
Bridge (a tab)	B	Section 10

Click the first point and click the second, or drag from one to the other. **Snap** pulls the cursor to corners, centers, midpoints and the drawing zero (within about 10 screen pixels, so it is more precise when zoomed in) and marks the spot in green. **Shift** constrains the shape (square, circle, angles in 45° steps, polygon rotation in 15° steps) and **Ctrl** switches snap off while you hold it. **Esc** cancels what you are doing; a second press goes back to *Select* and a third clears the selection. To pan without leaving the tool: middle button, or space bar + drag. *Draw* → *Draw with the mouse* → *Mouse options…* offers what the mouse can't give (rectangle corner radius, polygon sides, slot width, grid, snap, stroke color); these are not saved on exit.

With dialogs (*Draw* menu). *Draw shape* asks for the exact measurements of a rectangle (with corner radius), circle, ellipse, regular polygon (by radius or across flats), slot, and line or polyline (which can be closed). Shapes ask for a stroke color, which can't be that of a group. The remaining commands act on the selection:

Command	What it does
Move…, Place at…	A relative shift, or put the bottom-left corner of the selection at a given point
Rotate…, Resize…, Mirror…	About the center of the selection (or a point); resize from the center or a corner, with or without keeping proportions
Copy as array…, Copy around a circle…	Columns × rows with their spacing, or N shapes around a center

Command	What it does
Offset…	A parallel outline at a given distance, outward or inward, keeping the original or not
Union of shapes, Subtract shapes…, Intersection of shapes	Operations between closed shapes; in *Subtract shapes…* you choose which shape is the material
Close outline	Closes an open line

A hole left by a subtraction or an offset comes out as a separate shape inside the other one, which is how the application understands parts and holes.

Measurements of the selection. Below the tools there is a strip with X, Y, Width and Height (plus Diameter and Radius for a circle), which you can read and type into. X and Y refer to a point on the box that surrounds the shape, which you choose in the nine-point selector on the left (corners, side midpoints and center). **That point is the one that stays fixed when you resize.** With several shapes selected, the box is the one around all of them. The six buttons beside it align the selected shapes to each other (Align left, Center horizontally, Align right, Align top, Center vertically, Align bottom).

Editing. *Undo* (Ctrl+Z) and *Redo* (Ctrl+Y) go back up to 200 steps, also for groups and measurements; the history is not saved with the job. *Delete selection* is the Delete key. *Select the current board (with its frame)* takes a whole board so you can move or copy it complete. Edits are saved in the job as a list of operations that are replayed on the SVG; if the number of shapes in the SVG changes, they are discarded with a warning. A rectangle with something inside it is taken as a board: if it is really a part, untick *Detect boards*. An edited drawing can't be saved as an SVG.

10. Tabs

A tab is a piece of material that the tool doesn't cut, so the part stays attached to the board until you free it by hand. Tabs only work on through cuts, outside or inside (parts and holes, not open lines). You ask for them with the *Tabs* checkbox in the table row, or place them by hand.

Master switch. *Use tabs*, in the job settings. When it is off, the G-code comes out with no tabs at all (the ones from rows and the ones placed by hand stay stored and return when you switch it back on). Each PC remembers its setting, and so does the saved job.

Automatic proposal. With the checkbox ticked, the application proposes one tab for every 150 mm of perimeter (minimum two), evenly spaced and rotated until they fall on straight stretches, away from the corners. On a hole, the tab holds the cut-out piece in place. A part or hole that is too small is cut without tabs, and the window tells you.

Dimensions. The *width* is that of the tab that actually remains (6 mm): the tool lifts by that width plus its own diameter, so changing tools doesn't change the tab. The *height* is the material left above the bottom (2 mm): in 19 mm material, the tool lifts to Z −17. Without tabs, the G-code doesn't change by a single line. In the *Drawing* view, tabs show in amber over the outline (stronger for hand-placed ones) and the summary says how many there are.

By hand. With the tab tool (labeled *Bridge* in the tool strip, key B):

- Click on an outline to place a tab. Click on a tab to select it; drag it to slide it along the outline, pull one of its small handles to change its width, and press Delete to remove it (this doesn't delete the shape).

- The tab strip lets you type a tab's position, width and height; *Remove bridge* and *Shape to automatic* (or *Draw → Bridges of the chosen*) do what they say.
- When you touch a shape that has automatic tabs, its list switches to manual. Hand-placed tabs take priority over the row's checkbox.

Tabs **travel with the part**: moving, rotating, mirroring, resizing or copying the part carries its tabs along with it.

11. Hold-down and marks

The *Hold-down* tab helps you decide where to nail or screw the sheet to the machine bed without hitting a cut or a part, and it generates a short G-code that marks those points. When you open it, the window switches to the *Machine* view and paints in green the area where the sheet can be clamped, with the proposed points numbered.

Rules. They start from the real toolpath: the corridor the tool leaves, the edge of the sheet, and which pieces come loose when cut.

- Parts and interior cut-outs that come loose are never held down: only the frame around them is valid.
- There is a point on every large piece of scrap that comes loose, and just enough points so that no spot on the sheet is farther than the *Reach* from one. Points stay away from corners and cuts. On a large sheet they go in rows and columns, if they fit.
- If there is no other way to cover the sheet, the rules give way in this order: spacing, corners and, last of all, the clearance to the cut. A point closer to the cut than is comfortable is shown with a .

The numbers of the proposal, in the *Point proposal* box (an empty box takes the usual number):

Box	Default	What it is
Used	Nail	Nail-gun nail (2 mm head) or lag screw (up to 6 mm); changes both margins
Margin to the cut (mm)	6 (nail), 10 (lag screw)	Minimum distance from the cut edge to the point
Margin to the edge (mm)	12 (nail), 15 (lag screw)	Minimum distance from the edge of the sheet to the point
Mark height (mm)	5	Height of the tip above the sheet at each pause
Reach (mm)	300	No spot on the sheet is farther than this from a point
Minimum points	6	On a large sheet; on a small one, 3 if they fit
Minimum spacing (mm)	200	Never puts two points closer than this

There are also *Distance to corners (mm)* (100), *Distance to the cuts (mm)* (20), *In rows and columns* and *Bottom row at (mm)* (40). These numbers only apply to the proposal; by hand you can place a point anywhere inside the green area.

Adjusting. A click on the canvas adds a point (if one can't go there, the status bar says why); a click on a point removes it (or use the *Remove* button). You can type the coordinates in the table: a point in an unsafe spot is accepted, but it shows in red with the reason and the application asks before generating. *Propose again* discards the edits on that board. Each board has its own points; a drawing with no board has its own points on the assumed sheet (if the real sheet is larger, increase *Zero at (mm) from drawing*). They are saved with the job and checked again if the tool or the toolpath changes. Points are numbered from far to near, row by row, from the highest `Y` to the lowest.

Generating the marks. *Generate marks…* writes `NAME_board1_marks.nc` (or `NAME_marks.nc` if there is a single board). With the spindle stopped, it moves the machine to each point, rises to the mark height and pauses (`M0`). Load it into UGS; at each pause, mark the point under the tip and press *Resume*. At the end it does the same as the job (Home or return to origin, if ticked). Before nailing, move the machine's gantry out of the way.

12. Generating and saving

Generate. In the *Boards* list, tick the boards you want (if there are any) and press *Generate G-code…* (Ctrl+G). If something can't be calculated or doesn't fit the machine, the window says so and writes nothing. If everything is in order, it asks for the file name (with several boards, `_board1`, `_board2`… is added; if the file already exists, it asks before replacing it) and writes the file. A dialog and *Messages* report each file written, its number of lines, its minimum Z, the estimated time and the warnings. Then open the `.nc` in UGS.

Limits. Ruining material and tools is expensive; double-checking the data is cheap. These limits are the same for everyone: they are not bypassed, you correct the data.

Limit	Value	What happens when it is exceeded
Spoilboard margin	0.5 mm at most	Not calculated; the message states the limit
Material thickness	22 to 30 mm: alarm; over 30 mm: not generated	Alarm or block
Pass depth	At most 2 × the tool diameter (8 mm with the 4 mm tool). With no pass depth anywhere, the whole depth is a single pass	Not calculated; the message names the operation
Fits the machine	The toolpath must stay within the travel of the machine profile	Shown in the drawing but not written
G-code check	The generated G-code is reviewed: cuts above the surface, cuts without a feed, spindle off…	If anything is found, the file is not written

There is no way to know whether 8 mm at 1200 mm/min is too much for a hard material: the tool profile for that material decides that. The limit only blocks what makes no sense with any material.

Alarms (in red in *Messages* and in the summary): thickness over 22 mm, parts closer to each other than the tool diameter, areas the tool can't reach, and a toolpath that runs off the board (*Apply*

suggested shift brings it back inside). You don't have to confirm them one by one, but you do have to look at them before running the job.

Estimated time. The summary says how long the machine will be cutting (for example "≈ 1 h 35 min of machine time"), with the breakdown in the tooltip. It imitates the GRBL planner using the accelerations in the machine profile; if the profile doesn't include them, it calculates at constant speed, falls short, and warns you. It doesn't count marking, screwing down or cleaning.

The G-code. A plain-text `.nc` file, ASCII only.

Part	Contents
Header	Comments: material, margin, minimum Z, tool and machine
Startup	`G21 G90 G17 G94 G54`, rise to the safe Z, an `M0` pause if ticked, `M3 S20000` and a 2 s dwell
Cuts	`G0` and `G1`, with the feed (`F`) only when it changes
End	`M5`, Home (`$H`) or return to `X0 Y0` if ticked, your own end code, and `M30`

The `S20000` value is fixed and the machine doesn't interpret it: the actual RPM is set by hand on the VFD.

Saving the job. *File → Save job…* (Ctrl+S) writes a `.cncjob` with the drawing, the settings, the table, the groups, the edits, the tabs and the hold-down points; *Open saved job…* (Ctrl+Shift+O) brings it back. If the number of shapes in the SVG has changed, groups and edits are discarded and you are warned.

Job and G-code folders. *File → Job and G-code folders…* stores two paths: the jobs folder (`.cncjob`) and the G-code folder (`.nc`, including the marks G-code). The open and save dialogs start there, and you can pick a different folder from them without changing the fixed one. A path is valid if it is complete, exists and can be written to; if it stops being valid, the application warns you and asks what to do, and it never writes anywhere else without asking.

You want to	Use
Continue the same job another day	Saved job (`.cncjob`)
Apply the same split of operations to another drawing	Recipe (`.toml`)
Send the cut to the machine	The generated `.nc`, loaded into UGS

13. Shortcuts

Keyboard.

Shortcut	Command
Ctrl+O / Ctrl+Shift+O	Open SVG drawing / Open saved job
Ctrl+N	New empty drawing
Ctrl+S	Save job

Shortcut	Command
Ctrl+G	Generate G-code
Ctrl+T / Ctrl+M	Tool profiles / Machines
F5	Reload tool profiles
Ctrl+Z / Ctrl+Y	Undo / Redo
Ctrl+A / Esc / Delete	Select all / Clear selection / Delete selection
Ctrl+Shift+N / U / I	New group from the selection / Remove from group / Separate insides and outsides
Ctrl+1 / Ctrl+2 / Ctrl+0	*Drawing* view / *Machine* view / Fit view
V R C E P S L M B	Mouse tools (section 9)
Enter / Backspace	Finish the line / remove its last point
Ctrl+Q	Exit

Mouse on the canvas.

Action	What it does
Wheel	Zoom
Drag (*Select* tool), middle button, or space bar + drag	Pans the view
Double-click on the drawing	Fits the view (with a line half drawn, it finishes the line)
Click on a shape	Selects it and highlights its row
Ctrl+click / Shift+click	Add or remove / add only
Shift+drag	Box: selects what fits entirely inside
Click on empty space	Clears the selection
Right-click	Same as Esc; on a line with two or more points, it finishes the line
Shift / Ctrl (with a tool)	Constrain / switch snap off
Hold-down tab: click on a free spot / on a point	Adds / removes a point

Dragging an SVG onto the window opens it. Double-clicking a group name in the table renames it.